A Thermo-Optical Convection Layer above an OLED Display as a Programmable Analog Medium

Svetoch Series, Paper III of VI

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Abstract

The companion papers in this series route light from a smartphone's OLED screen to its front camera with an external flat mirror. Here we remove the mirror and ask whether the *air itself* can close the loop. With the phone lying screen-up, the heated OLED warms a thin layer of air above the cover glass; buoyancy drives a laminar natural-convection boundary layer whose temperature — and hence refractive-index — field is set, pixel by pixel, by what the screen displays. The air becomes an unmodified, programmable phase spatial light modulator (SLM), written by the OLED and read by the front camera through Background-Oriented Schlieren (BOS). From first principles, a Rayleigh number $Ra \approx 6.2 \times 10^6$ (laminar, $Ra < 10^9$) yields a thermal-layer thickness $\delta_T \approx L\,Ra^{-1/5} \approx 6.9\,\text{mm}$ matching the screen-to-camera gap; the Edlén relation $\mathrm{d}n/\mathrm{d}T \approx -9.4 \times 10^{-7}\,\mathrm{K}^{-1}$ turns a $\Delta T = 20\,°\mathrm{C}$ heating into $\Delta n \approx -1.8 \times 10^{-5}$; a $\Lambda = 4\,\text{mm}$ thermal grating deflects rays by $\theta_x \approx -6.6 \times 10^{-5}\,\text{rad}$ ($\sim 0.41\,\text{px}$, sub-pixel, BOS-recoverable); and the optical-path difference (OPD $\approx 126\,\text{nm}$) gives a $\sim 85°$ phase shift and visible Fabry–Pérot moiré. The demonstrated effect is a **low-SNR (≈ 1.3) sub-pixel deflection**, with controlled shifts up to ~ 1.6–2.0 equivalent pixels under $\times 15$ upscaling. We outline applications — resonant aerosol actuation, mobile schlieren/leak imaging, and a speculative "air-as-coprocessor" matrix engine — and state plainly which are demonstrated and which remain conjectural pending higher SNR.

Keywords: Background-Oriented Schlieren, natural convection, Rayleigh number, refractive index, Edlén equation, phase SLM, OLED, thermo-optics, smartphone sensing, analog computing.

1 Introduction

Paper I showed that an OLED screen, an inexpensive flat mirror, and a front camera form a self-contained analog matrix engine. The mirror is the one component not already in the phone. This paper asks whether it can be eliminated by letting a physical field in the air — rather than a reflective surface — carry the optical signal back to the camera.

A phone lying screen-up is a warm horizontal plate. Driving the OLED bright raises the cover glass a few degrees above ambient; the air above is heated and rises by buoyancy, forming a natural-convection thermal boundary layer. Since the refractive index of air depends weakly but measurably on temperature, the displayed brightness pattern is *written* into the air as a refractive-index pattern. Light skimming through the layer is deflected, and the front camera reads the deflection. Waste heat becomes the medium itself.

This is the **mirror-less channel**: screen → heated air boundary layer → camera, with no added optics (Figure 1). The air acts as a programmable phase SLM; the read-out is Background-Oriented Schlieren (BOS), in which a structured reference background, distorted by index gradients, is recovered by image cross-correlation — and here the reference background is painted by the screen itself.

We state the scientific status at the outset. The original derivation was *inspired by* a structural analogy to a speculative "Vacuum Mass Fraction / Null-Vector Gravity" (VMF/NVG) picture of the QCD vacuum, mapping a condensate amplitude and Goldstone phase onto the temperature and flow-potential fields. We treat that analogy strictly as mathematical scaffolding and engineering inspiration, and make *no* claim about real vacuum or QCD physics. The actual basis is uncontroversial physics: Navier–Stokes natural convection, the Rayleigh number, the Edlén index relation, schlieren/BOS ray optics, and Fabry–Pérot interference.

Figure 1. The OLED–mirror–camera optical channel (side view)

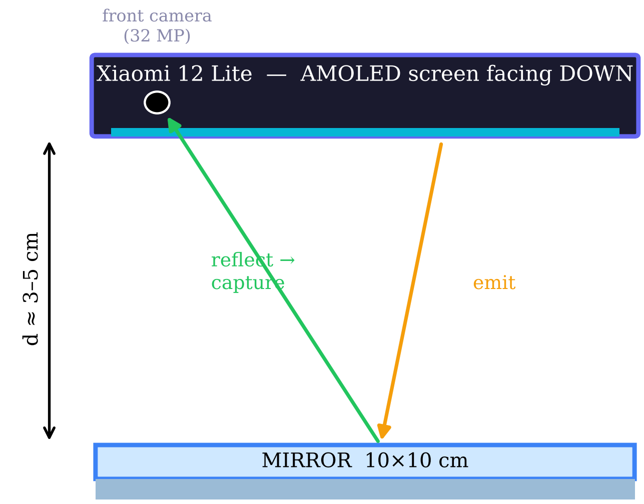

Figure 1: Device geometry. For the mirror-less channel the phone lies screen-up; the heated OLED warms the air boundary layer above the cover glass, which the front camera reads through the same few-millimetre gap.

2 Theory of the thermal boundary layer

2.1 Convection regime and layer thickness

For a heated horizontal plate the regime is set by the Rayleigh number

$$Ra = \frac{g\,\beta\,\Delta T\,L^3}{\nu\,\alpha}, \tag{1}$$

with $g = 9.81\,\mathrm{m\,s^{-2}}$, $\beta = 1/T_0 \approx 3.3\times10^{-3}\,\mathrm{K^{-1}}$, $\nu \approx 1.6\times10^{-5}\,\mathrm{m^2\,s^{-1}}$, $\alpha \approx 2.2\times10^{-5}\,\mathrm{m^2\,s^{-1}}$, $L \approx 0.15\,\mathrm{m}$ (a 6.55″ display), and a full-white heating $\Delta T = 20\,^\circ\mathrm{C}$. Then

$$Ra \approx \frac{9.81\times3.3\times10^{-3}\times20\times0.15^3}{1.6\times10^{-5}\times2.2\times10^{-5}} \approx 6.2\times10^6. \tag{2}$$

Since $Ra < 10^9$, the flow is firmly **laminar**, forming a stable convection layer. From boundary-layer similarity,

$$\delta_T \approx L\,Ra^{-1/5} \approx 0.15\times(6.2\times10^6)^{-0.2} \approx 6.9\ \mathrm{mm}, \tag{3}$$

which matches the optical path from the OLED to the front-camera pupil through the cover glass, $d_{\mathrm{gap}} \approx 5$–$7\,\mathrm{mm}$: the camera light traverses essentially the full depth of the programmable layer (Figure 2).

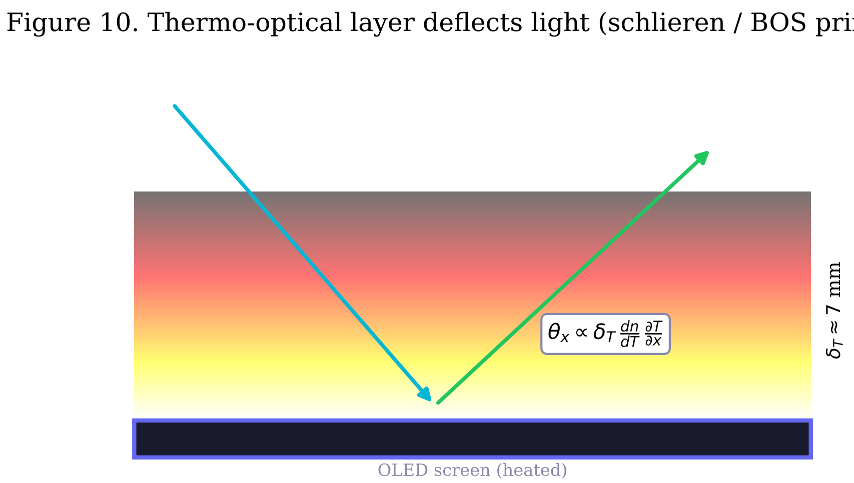

Figure 2: The thermo-optical layer. Buoyancy above the heated OLED forms a laminar boundary layer of thickness $\delta_T \approx 7\,\mathrm{mm}$; rays crossing its lateral index gradient are deflected by $\theta_x \approx \delta_T\,(\mathrm{d}n/\mathrm{d}T)\,(\partial T/\partial x)$ and read as a sub-pixel shift (BOS).

2.2 Refractive-index modulation (Edlén)

The temperature dependence of the index of air (Edlén, visible, standard pressure) is

$$\frac{\mathrm{d}n}{\mathrm{d}T} \approx -9.4 \times 10^{-7}\ \mathrm{K}^{-1}, \tag{4}$$

so a $\Delta T = 20\,^{\circ}\mathrm{C}$ contrast writes an index contrast

$$\Delta n = \frac{\mathrm{d}n}{\mathrm{d}T}\,\Delta T \approx -1.8 \times 10^{-5}. \tag{5}$$

Small in absolute terms, but enough for a recoverable signal because both deflection and phase accumulate over the millimetre-scale layer.

2.3 Schlieren ray deflection

A ray crossing a lateral index gradient over the layer depth is deflected by

$$\theta_x = \int_0^{\delta_T} \frac{1}{n}\frac{\partial n}{\partial x}\,\mathrm{d}z \approx \delta_T\,\frac{\mathrm{d}n}{\mathrm{d}T}\,\frac{\partial T}{\partial x}. \tag{6}$$

For a displayed thermal grating of period $\Lambda \approx 4\,\mathrm{mm}$ (Figure 3), the edge gradient is

$$\frac{\partial T}{\partial x} \approx \frac{\Delta T}{\Lambda/2} \approx \frac{20\ \mathrm{K}}{2 \times 10^{-3}\ \mathrm{m}} \approx 10^{4}\ \mathrm{K\,m^{-1}}, \tag{7}$$

giving

$$\theta_x \approx (7 \times 10^{-3}\,\mathrm{m})(-9.4 \times 10^{-7}\,\mathrm{K}^{-1})(10^{4}\,\mathrm{K\,m^{-1}}) \approx -6.6 \times 10^{-5}\ \mathrm{rad}. \tag{8}$$

Over the gap this displaces the imaged background by

$$\Delta x_{\mathrm{sensor}} = d_{\mathrm{gap}}\,\theta_x \approx 5 \times 10^{-3}\,\mathrm{m} \times 6.6 \times 10^{-5}\,\mathrm{rad} \approx 0.33\ \mu\mathrm{m}, \tag{9}$$

which, at the 0.8 μm front-camera pitch, is

$$\Delta x_{\mathrm{px}} \approx \frac{0.33\ \mu\mathrm{m}}{0.8\ \mu\mathrm{m/px}} \approx 0.41\ \mathrm{px}. \tag{10}$$

A ~ 0.4 px shift is sub-pixel but well within BOS cross-correlation, which routinely resolves ~ 0.05 px.

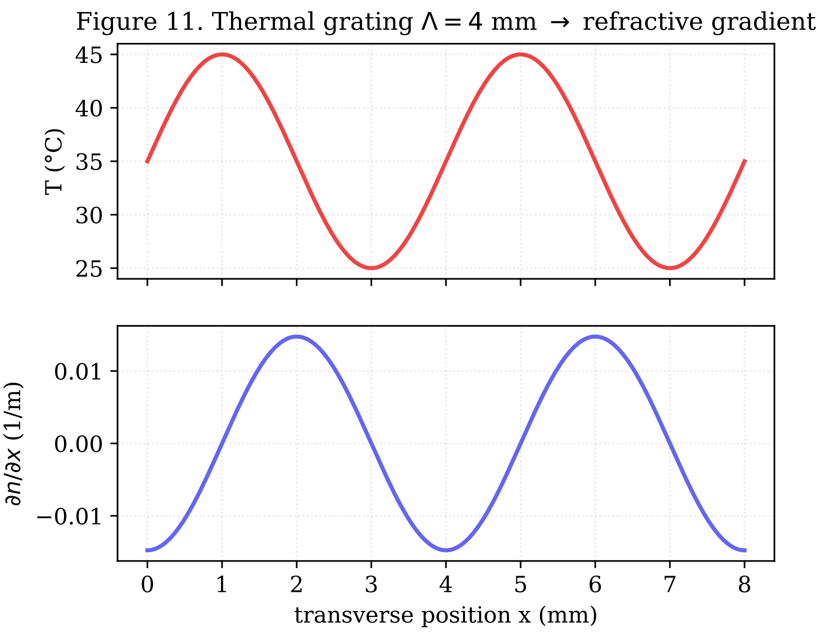

Figure 3: A $\Lambda = 4$ mm thermal grating on the OLED. The temperature profile $T(x)$ and the resulting lateral index gradient $\partial n/\partial x$ define the deflection field read by the camera.

2.4 Fabry–Pérot moiré: the layer as a phase SLM

The index contrast also accumulates phase along the depth. The optical-path difference between a hot and a cold column is

$$\mathrm{OPD} = |\Delta n|\, \delta_T \approx 1.8 \times 10^{-5} \times 7 \times 10^{-3}\,\mathrm{m} \approx 126\ \mathrm{nm}. \tag{11}$$

For the green OLED sub-pixel, $\lambda = 532$ nm, this is

$$\varphi = 2\pi\,\frac{\mathrm{OPD}}{\lambda} = 2\pi\,\frac{126}{532} \approx 1.49\ \mathrm{rad} \approx 85^\circ. \tag{12}$$

An 85° phase shift is large; modulating the multiple-beam Fabry–Pérot interference in the cover glass, it produces strong spatial bands — the moiré visible above the glass. Hence the central claim: the thermal layer is a **programmable phase spatial light modulator**, its phase map set by displayed brightness. The screen writes phase into the air; the camera reads it.

3 Read-out by Background-Oriented Schlieren

BOS reconstructs an index-gradient field by comparing a reference frame and a perturbed frame through local cross-correlation (or block matching, e.g. sum-of-absolute-differences) of a structured background. The pixel displacement $\Delta x(x, y)$ is proportional to the line-integrated index gradient — exactly the θ_x above.

The mirror-less channel supplies its own background: the OLED paints a high-frequency reference (grating or speckle), captures a cold baseline frame, then heats and captures the perturbed frame. No external grid, collimator, or knife-edge is needed — the structured background and the heat source are the same device.

Measured performance. The experiment recorded controlled displacements up to ~ 1.6–2.0 *equivalent* pixels, where the equivalence includes $\times 15$ digital upscaling (~ 0.1–0.13 physical pixels), consistent with the $\sim 0.4\,\mathrm{px}$ first-principles estimate to within the model's crudeness. The recovered shift had SNR ≈ 1.3: real but **weak**, sitting close to the floor set by shot noise, focus breathing, ambient air currents, and device drift. SNR is the governing figure of merit for every application below, and raising it is the single most important open problem.

4 Applications

4.1 Resonant thermo-optical actuation of aerosols (strong)

Displaying a large quad-vortex pattern and modulating its intensity at the convection resonance ($\approx 5\,\mathrm{Hz}$) drives a macroscopic thermal column. With smoke or aerosol present, the resonantly amplified flow entrains particles into a visible macro-vortex, which the same camera quantifies by optical flow. Being a large, directly imaged motion rather than a sub-pixel deflection, this is the most robustly demonstrated application and the least dependent on BOS SNR.

4.2 Mobile schlieren / BOS visualizer (strong)

Section 3 is itself an instrument: a pocket schlieren/BOS visualizer using the screen's own reference pattern. It images thermal plumes, convection, and gas leaks by detecting index disturbances above a 3σ threshold and classifying them by spectral-temporal signature. The novelty is using the device's own display as the structured BOS background, removing the bench optics schlieren normally requires.

4.3 "Air as coprocessor": thermo-optical matrix multiply (speculative)

If weights are encoded as a displayed brightness pattern, the induced index gradients implement a spatial multiply and the BOS-measured deflection integrates a vector–matrix product; eight heated zones could act as a parallel bus, and vortex chirality could encode an activation sign. This "Thermal LLM" is attractive but **gated by SNR**: at SNR ≈ 1.3 per zone the arithmetic accuracy is far below a useful accelerator's. It is *promising but speculative pending substantially higher SNR* (e.g. acousto-optic high-frequency thermal modulation, sharper $\partial n/\partial x$ from fine checkerboard masks, or graphene-assisted heat localization).

4.4 Further directions (speculative)

Three further concepts follow from the same physics, reported as conjectures: (B4) computing **topological invariants** (Gauss linking numbers) from the optical-flow field of interacting thermal vortices, with hysteresis as medium memory; (B7) a **volumetric aerosol display**, rendering density structures visible in scattered light, with a dry control frame separating real structure from artifacts; and (B8) a **contactless thermo-optical router**, switching a thermal lens/vortex between channels by addressing heated zones. Each is novel in modality but has uncertain enablement at present SNR.

5 Methods

Hardware: Xiaomi 12 Lite (6.55″ AMOLED, 2400 × 1080, 120 Hz, OLED pitch 63.2 µm; 32 MP front camera, photosite pitch ≈0.8 µm, min. focus ≈10 cm). For the mirror-less channel the phone lies *screen-up* on a rigid stand in still air; the path is screen → cover glass → heated air layer → front-camera pupil, gap $d_{\text{gap}} \approx 5$–7 mm. **Protocol:** (1) display a high-frequency reference ($\Lambda \approx 4$ mm grating, or speckle) and capture a cold baseline; (2) drive the heating/grating pattern, letting the layer develop over $\tau_{\text{thermal}} \approx 100$–$200$ ms; (3) capture the perturbed frame; (4) recover the displacement by BOS cross-correlation (block matching, ×15 upscaling) and report shift and SNR. For actuation, modulate a quad-vortex at ≈ 5 Hz and quantify entrained flow by optical flow. **Software:** as in Paper I, a dependency-free Python HTTPS server relays Start/Stop and stores runs as JSON; browser ES modules render patterns, capture frames via `getUserMedia`, compute the BOS metric, and post results. **Figures:** theory figures plot the closed-form relations of Section 2; produced by `papers/scripts/make_figures.py`.

6 Discussion and limitations

The effect is real but weak. What is demonstrated is a controlled, repeatable, sub-pixel deflection produced by displayed heat and recovered by BOS, SNR ≈ 1.3 — enough to *visualize* index gradients (4.1–4.2) but not yet to *compute* reliably (4.3–4.4). The matrix-multiply, router, and volumetric-display applications are engineering conjectures whose enablement hinges on raising SNR by one to two orders of magnitude.

Patentability framing. Our internal review rates the actuation (A1) and the mobile schlieren/BOS visualizer (A6) as strong — demonstrable effects with narrow claims — while the thermal matrix multiply (B5), volumetric display (B7), and router (B8) are moderate, with enablement risk driven by the low SNR.

The VMF/NVG analogy is an analogy. The condensate-amplitude ↔ temperature and Goldstone-phase ↔ flow-potential correspondence is a structural mnemonic only. Nothing here tests, supports, or depends on any claim about the QCD vacuum; the physics used stands on its own established footing.

Other limitations. The thermal time constant (∼100–200 ms) caps the native modulation rate; ambient currents, focus breathing, and device drift degrade SNR; and the close $\delta_T \approx d_{\text{gap}}$ coincidence is device-specific and should be re-derived per phone.

7 Conclusion

Removing the mirror, the air above a heated OLED becomes a programmable phase SLM: the screen writes an index pattern into a laminar convection layer ($Ra \approx 6.2 \times 10^6$, $\delta_T \approx 6.9$ mm), the front camera reads the sub-pixel deflection ($\theta_x \approx -6.6 \times 10^{-5}$ rad) by BOS, and a $\sim 85°$ Fabry–Pérot phase shift makes it visible as moiré. The channel is genuine but low-SNR (≈ 1.3); it already enables resonant aerosol actuation and mobile schlieren imaging, while the compute, router, and display applications remain promising conjectures pending higher SNR.

Data and code. github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vm (Apache-2.0).

Priority note. Released as a defensive publication to establish authorship and date of disclosure; the author has elected not to seek patent protection.

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Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.